Topology: MAT307/M405/MAT803 : - Problem Set 5

1. Prove that the punctured euclidean space $X = \mathbb{R}^n \backslash \{\mathbf{0}\}$ is path connected if and only if $n > 1$.

2. Prove that $\mathbb{R}^2 \setminus \mathbb{Q}^2$ with the subspace topology of $\mathbb{R}^2$ is connected, but $\mathbb{R} \setminus \mathbb{Q}$ with the subspace topology of $\mathbb{R}$ is not connected.

3. Let X and Y be topological spaces, and suppose there exist embeddings $f : X \to Y$ and $g : Y \to X$. Is it necessary that X and Y are homeomorphic? Prove or provide a counterexample.

4. Let X be a non-empty topological space and let $\mathcal{B}$ be a base for the topology on X. Prove that X is compact if and only if every open cover of X consisting of elements of $\mathcal{B}$ has a finite subcover.

5. Let $\mathcal{T}_1$ and $\mathcal{T}_2$ be two topologies on X such that both $(X, \mathcal{T}_1)$ and $(X, \mathcal{T}_2)$ are compact Hausdorff spaces. Prove that either $\mathcal{T}_1 = \mathcal{T}_2$ or they are not comparable.

6. Prove that every compact subspace of a metric space is closed and bounded in that metric topology. Find an example of a metric space in which a closed and bounded subspace is not compact.

7. Let X be a compact metric space. Prove that if $f : X \to X$ is an isometry (distance preserving map) of X, then f is a homeomorphism.

8. Let $\mathbb{R}$ be given the cocountable topology. Is the closed interval $[0, 1]$ a compact subspace of $\mathbb{R}$ under this topology?

9. Let $f : X \to Y$ be a continuous map between topological spaces. Show that if X is compact and Y is Hausdorff, then f is a closed map.

10. Prove that if Y is compact, then the projection $\pi_X : X \times Y \to X$ is a closed map.

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